

English	(1015CMD303361230017)		Test Pattern
		CLASSROOM CONTACT PROGRAMME	NEET (UG)
	(Academic Session : 2023-2024)		MINOR
			21-05-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MED-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill 0012345678.

Test Booklet Code

This Booklet contains 28 pages.

E3

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in Pre-Medical 2024

Topic : RAY OPTICS (TILL DATE) FROM TYPES OF MAGNIFICATION, CURRENT ELECTRICITY (TILL DATE) [BACK UNIT] : NLM, FRICTION AND GRAVITATION

TG: @Chalnaayaaar

SECTION-A (PHYSICS)

1. As we go from the equator to the poles, the value of g :

- (1) Remains the same
- (2) Decreases
- (3) Increases
- (4) Decreases up to a latitude of 45° and then increases

2. Mass particles of 1 kg each are placed along x-axis at $x = 1, 2, 4, 8, \dots, \infty$. Then gravitational force on a mass of 3 kg placed at origin is (G = Universal gravitational constant):-

- (1) $4G$
- (2) $\frac{4G}{3}$
- (3) $2G$
- (4) ∞

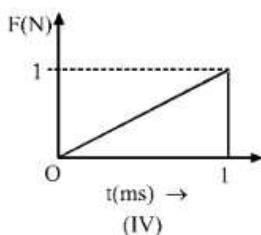
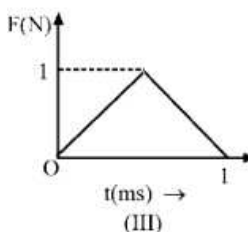
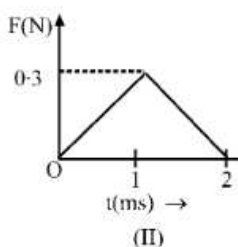
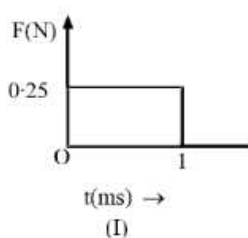
3. Mass density of a solid sphere is ρ . Radius of the sphere is R . The gravitational field at a distance r from the centre of the sphere inside it is-

- (1) $\frac{4\rho G\pi r}{3}$
- (2) $\frac{4\rho G r^2}{3}$
- (3) $\frac{4\rho G R^3}{3r^2}$
- (4) $\frac{\rho G R^3}{\pi r}$

4. If momentum of a particle is given by $P = (4t^2 - 2t)$ N.s then find the force acting on it, at $t = 1$ sec :-

- (1) 8 N
- (2) 2 N
- (3) 6 N
- (4) - 2 N

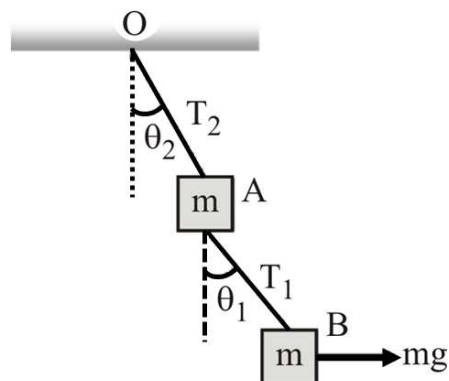
5. Figures I, II, III and IV depicts variation of force with time



In which situation impulse will be maximum

- (1) I & II
- (2) III & I
- (3) III & IV
- (4) Only IV

6. In the following figure the masses of the blocks A and B are same and each equal to m . The tensions in the strings OA and AB are T_2 and T_1 respectively. The system is in equilibrium with a constant horizontal force mg on B. The T_1 is

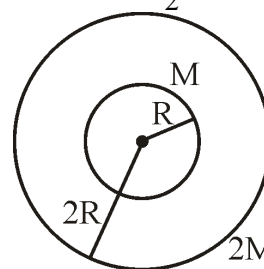


- (1) mg
- (2) $\sqrt{2}mg$
- (3) $\sqrt{3}mg$
- (4) $\sqrt{5}mg$

7. A missile is launched with a velocity less than the escape velocity. The sum of its kinetic energy and potential energy will be -

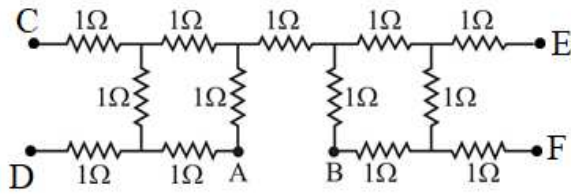
- (1) Positive
- (2) Negative
- (3) Negative or positive uncertain
- (4) zero

8. Two concentric hollow spheres of mass $2M$ and M having radius $2R$ and R respectively are placed as shown in figure. The gravitational potential at a distance of $\frac{3R}{2}$ from the centre is-



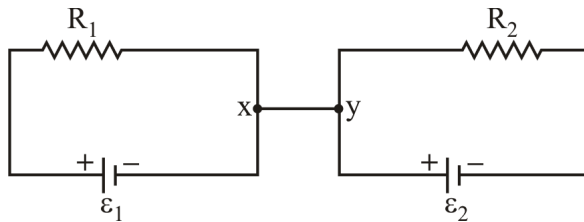
- (1) $-\frac{GM}{R}$
- (2) $-\frac{2GM}{3R}$
- (3) $-\frac{5GM}{3R}$
- (4) $-\frac{GM}{3R}$

20. In the network of resistances as shown in figure, the effective resistance between points A and B is :-



- (1) $\frac{5\Omega}{2}$ (2) $\frac{3\Omega}{2}$ (3) $\frac{3\Omega}{4}$ (4) None

21. In the circuit shown below, the conductor XY is of negligible resistance then :-



- (1) Current will flow through xy if $\varepsilon_1 \neq \varepsilon_2$
 (2) Current will flow through xy if $\frac{\varepsilon_1}{R_1} \neq \frac{\varepsilon_2}{R_2}$

Current will flow through xy if

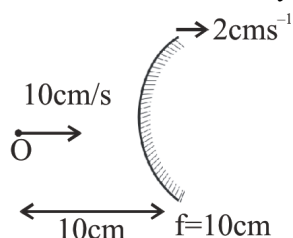
- (3) $\frac{\varepsilon_1 + \varepsilon_2}{R_1 + R_2} \neq \frac{|\varepsilon_1 - \varepsilon_2|}{R_1 - R_2}$

- (4) No current will flow through x-y

22. If a metal wire radius r is decreased by stretching it then its resistivity.

- (1) Constant
 (2) is proportional to $\frac{1}{r^2}$
 (3) is proportional to $\frac{1}{r^4}$
 (4) is proportional to $\frac{1}{r^3}$

23. What is the velocity of image in situation shown in figure. Object moves with velocity 10 cm/s and mirror moves with velocity 2 cm/s as shown.



- (1) 2 cm/s (2) 8 cm/s (3) 5 cm/s (4) 0

24. A beam of monochromatic light travels in a medium of refractive index μ from vacuum, then the value of wavelength of wave in the medium will be :-

- (1) $(\lambda_{\text{vacuum}})\mu$ (2) $(1/\lambda_{\text{vacuum}})\mu$
 (3) $(\mu/\lambda_{\text{vacuum}})$ (4) $(\lambda_{\text{vacuum}}/\mu)$

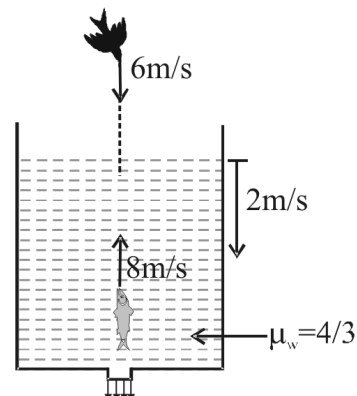
25. A light ray enter in a glass slab of R.I. = $\sqrt{3}$. The angle of incident is 60° . If ray travel 5 cm inside the slab and emerge out of the slab. Find the lateral shift between the incident ray and emergent ray :-

- (1) $\frac{5}{2}$ cm (2) $\frac{7}{2}$ cm
 (3) $\frac{5\sqrt{3}}{2}$ cm (4) $\frac{5}{\sqrt{2}}$ cm

26. An under water swimmer is at a depth of 12 m below the surface of water. A bird is at height of 18 m from the surface of water directly above his eyes. For swimmer the bird appears to be at a distance from the surface of water equal to (Refractive index of water = $4/3$) :-

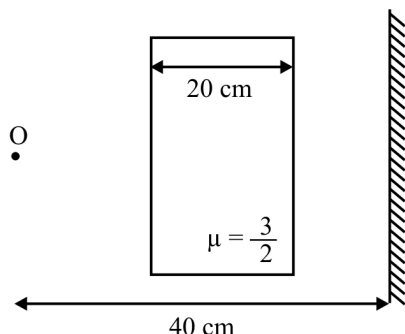
- (1) 24 cm (2) 12 cm
 (3) 18 cm (4) 9 m

27. As shown in the figure bird is moving downward with a speed of 6m/s and fish is moving upward with a speed of 8m/sec. At the instant shown the water level falls at the rate of 2m/sec. Velocity of fish as seen by bird is.

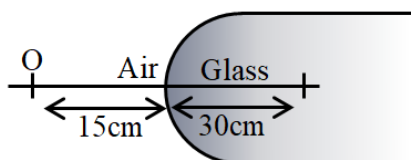


- (1) $\frac{11}{2} \hat{j}$ (2) $\frac{23}{2} \hat{j}$
 (3) $\frac{15}{2} \hat{j}$ (4) Zero

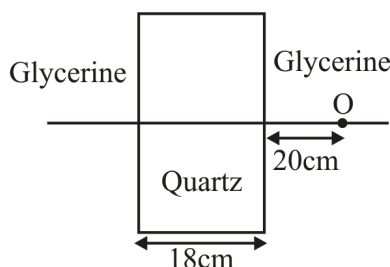
28. A 20 cm thick glass slab of refractive index 1.5 is kept in front of a plane mirror. Find the position of the image (relative to mirror) as seen by an observer through the glass slab when a point object is kept in air at a distance of 40 cm from the mirror.



- (1) $\frac{110}{3}$ (2) $\frac{100}{3}$ (3) $\frac{80}{3}$ (4) $\frac{50}{3}$
29. A prism is made up of a material of refractive index $\sqrt{3}$. The angle of prism is A. If the angle of minimum deviation is equal to the angle of the prism then the value of A
- (1) 30° (2) 60° (3) 45° (4) 75°
30. A point object O is placed in front of a glass rod having spherical end of radius of curvature 30 cm. The image would be formed



- (1) 30 cm left (2) infinity
(3) 1 cm to the right (4) 18 cm to the left
31. Given that velocity of light in quartz is 3×10^6 m/s & velocity of light in glycerine is $\frac{27}{2} \times 10^6$ m/s. Now a slab made of quartz is placed in glycerine as shown. What is the shift produced by slab as seen from other side.



- (1) 12 cm (2) 14 cm (3) 6 cm (4) 9 cm

32. A microscope is focussed on a mark on a piece of paper & then a slab of glass of thickness 9 cm & refractive index 1.5 is placed over the mark. How should the microscope be moved to get the mark in focus again :-

- (1) 1 cm upward
(2) 3 cm downward
(3) 3 cm upward
(4) 2 cm upward

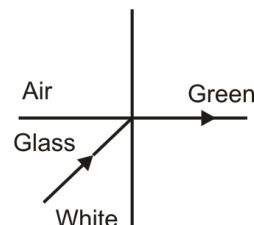
33. A thin prism having refracting angle 10° is made of glass of refractive index 1.42. This prism is combined with another thin prism of glass of refractive index 1.7. This combination produces dispersion without deviation. The refracting angle of second prism is :-

- (1) 6° (2) 8°
(3) 10° (4) 4°

34. A vessel is half filled with a liquid of refractive index μ . The other half of the vessel is filled with an immiscible liquid of refractive index 2μ . The apparent depth of the vessel is 25% of actual depth. The value of μ is :-

- (1) $\frac{1}{3}$ (2) $\frac{3}{1}$
(3) $\frac{4}{3}$ (4) $\frac{3}{4}$

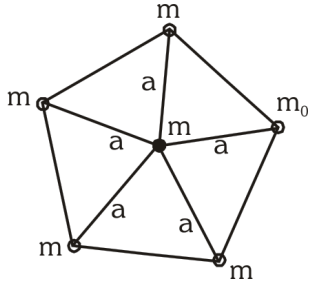
35. White light is incident on the interface of glass and air as shown in the figure. If green light is just totally internally reflected then the reflected ray contains



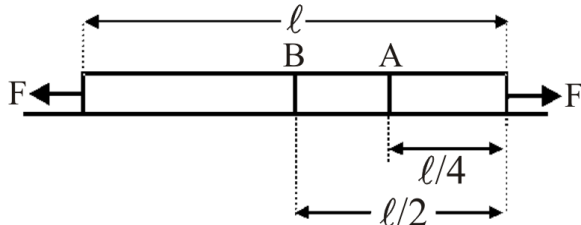
- (1) Yellow, orange, red
(2) Violet, indigo, blue
(3) All colours
(4) All colours except green

SECTION-B (PHYSICS)

36. Five bodies of mass m, m, m, m and m_0 are placed at corners of a pentagon as shown in figure. The net gravitational force on body of mass ' m ' placed at centre is—



- (1) Zero
 (2) $\frac{Gmm_0}{a^2}$
 (3) $\frac{Gm(m_0 + m)}{a^2}$
 (4) $\frac{Gm(m_0 - m)}{a^2}$
37. A heavy rope of mass m and length ℓ is placed on a smooth table. It is pulled at both ends by applying force F of equal magnitude as shown. Tension in the rope at point



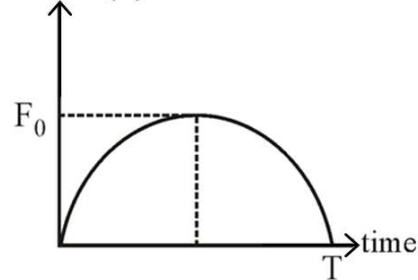
- (1) A is $3F/4$ (2) B is $F/2$
 (3) A is F (4) B is $2F$
38. An artificial satellite is moving in a circular orbit around the earth. The height of the satellite above the surface of earth is R . Suppose the satellite is stopped suddenly in its orbit and allowed to fall freely, on reaching earth, its speed will be :- (g is acceleration due to gravity at earth's surface, R is Radius of earth)

- (1) $\sqrt{gR}$ (2) $2\sqrt{gR}$
 (3) $3\sqrt{gR}$ (4) $5\sqrt{gR}$

39. The potential energy of a body of mass 3 kg on the surface of a planet is -54 Joule. The escape velocity from the surface of this planet is:-

- (1) 18 m/s (2) 162 m/s
 (3) 36 m/s (4) 6 m/s

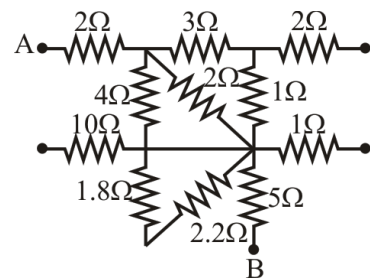
40. Force (F)



A particle of mass m , initially at rest is acted upon by a variable force F for a brief interval of time T . It begins to move with a velocity U after the force stops acting. F is shown in the graph as a function of time, then the value of U is:-

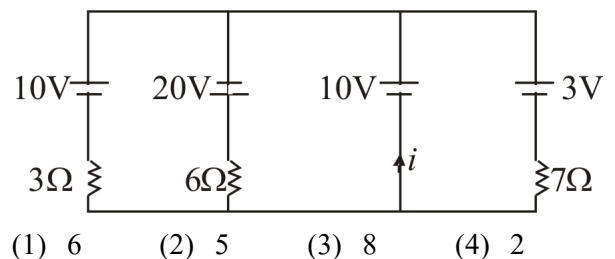
- (1) $U = \frac{\pi F_0^2}{2m}$ (2) $U = \frac{\pi F_0 T}{4m}$
 (3) $U = \frac{\pi T^2}{8m}$ (4) $U = \frac{F_0 T}{2m}$

41. What is the equivalent resistance between the point A and B of the network :-



- (1) $\frac{57}{7} \Omega$ (2) 8Ω
 (3) 6Ω (4) $\frac{57}{5} \Omega$

42. In the given figure, all the batteries are ideal. Find the value of i (in amperes)

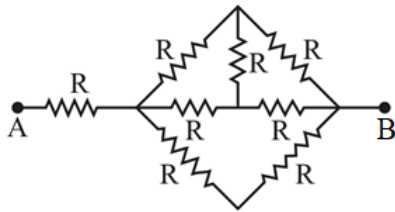


- (1) 6 (2) 5 (3) 8 (4) 2

43. Two wires of equal diameters, resistivity ρ_1 and ρ_2 and lengths x_1 and x_2 respectively are joined in series, then the equivalent resistivity of the combination is :-

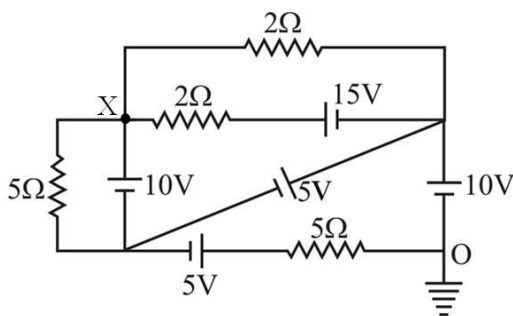
(1) $\frac{\rho_1 x_1 + \rho_2 x_2}{x_1 + x_2}$ (2) $\frac{\rho_1 x_2 - \rho_2 x_1}{x_1 - x_2}$
 (3) $\frac{\rho_1 x_1 + \rho_2 x_1}{x_1 + x_2}$ (4) $\frac{\rho_1 x_1 + \rho_2 x_2}{\rho_1 \rho_2}$

44. In the network of resistances as shown in figure, the effective resistance between points A and B is :-



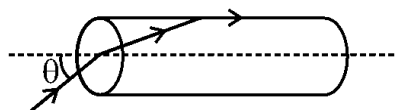
(1) $8R$ (2) $5R$ (3) $\frac{5}{3}R$ (4) $\frac{8}{3}R$

45. In the circuit shown if point O is earthed, the potential of point X is equal to :-



(1) 10 V (2) 15 V
 (3) 25 V (4) 12.5 V

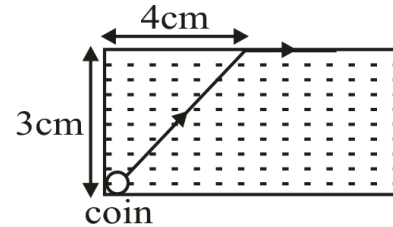
46. A transparent solid cylindrical rod has a refractive index of $\frac{2}{\sqrt{3}}$. It is surrounded by air. A light ray is incident at the mid-point of one end of the rod as shown in the figure :-



The incident angle θ for which the light ray grazes along the wall of the rod is

(1) $\sin^{-1}(1/2)$ (2) $\sin^{-1}(\sqrt{3}/2)$
 (3) $\sin^{-1}(2/\sqrt{3})$ (4) $\sin^{-1}(1/\sqrt{3})$

47. A small coin is resting on the bottom of a beaker filled with a liquid. A ray of light from coin travels upto the surface of the liquid & moves along its surface. How fast is the light travelling in liquid.

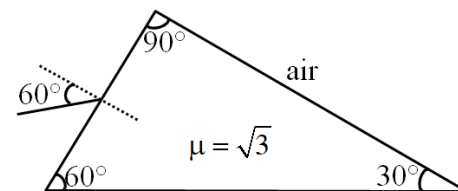


(1) 1.2×10^8 m/s
 (2) 1.8×10^8 m/s
 (3) 2.4×10^8 m/s
 (4) 3.0×10^8 m/s

48. A ray of light is incident in oil from air. It's reflected part & refracted part are mutually perpendicular. Find R.I. of oil while angle of incidence is 30° .

(1) $\sqrt{3}$ (2) $\frac{2}{\sqrt{3}}$
 (3) $\frac{1}{\sqrt{3}}$ (4) None

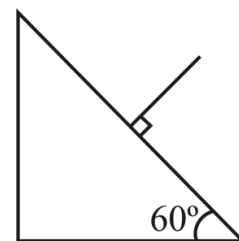
- 49.



Total deviation of the ray for a prism placed in air as shown in diagram $\left(\sin 35^\circ = \frac{1}{\sqrt{3}} \right)$

(1) 30° (2) 60° (3) 120° (4) 240°

50. Find the value of emergence from the prism. R.I. of the glass is $\sqrt{3}$.



(1) 60° (2) 30° (3) 45° (4) 90°

Topic : GOC-I, ELECTROPHILIC ADDITION REACTION, ELECTROCHEMISTRY (TILL SALT BRIDGE), SURFACE CHEMISTRY [BACK UNIT] : CHEMICAL KINETICS

TG: @Chalnaayaaar

SECTION-A (CHEMISTRY)

51. x g is the amount of a gas adsorbed by m g of a solid. If multilayer adsorption is supposed to occur, then which of the following statement is incorrect :-

- (1) At low pressure, (x/m) increases linearly with gas pressure.
- (2) At low pressure, (x/m) increases less than proportionally to gas pressure.
- (3) At moderate pressure, (x/m) does not increase linearly with gas pressure.
- (4) At high pressure, (x/m) becomes independent of gas pressure.

52. On adding AgNO_3 solution into KI solution, a negatively charged colloidal sol is obtained when they are in :-

- (1) 50 mL of 0.1 M AgNO_3 + 50 mL of 0.1 M KI
- (2) 50 mL of 0.1 M AgNO_3 + 50 mL of 0.2 M KI
- (3) 50 mL of 0.2 M AgNO_3 + 50 mL of 0.1 M KI
- (4) None of these

53. Which of the following statement is correct?

- (1) Greater is the protective power of lyophilic colloid, lesser is its gold number.
- (2) Greater is the protective power of lyophilic colloid, higher is its gold number.
- (3) Protective power of lyophilic colloid is not measured by gold number.
- (4) None of these

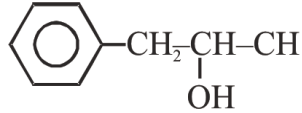
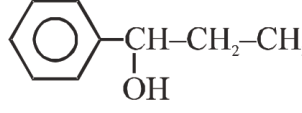
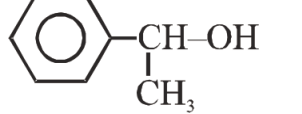
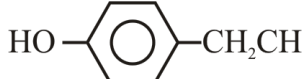
54. In which of the following reactions, heterogeneous catalysis takes place :

- (a) $\text{CH}_3\text{COOC}_2\text{H}_5(\ell) + \text{H}_2\text{O}(\ell) \xrightarrow{\text{HCl}(\ell)}$
 $\text{CH}_3\text{COOH}(\text{aq}) + \text{CH}_3\text{OH}(\text{aq})$
 - (b) Vegetable oil (ℓ) + $\text{H}_2(\text{g}) \xrightarrow{\text{Ni}(\text{s})}$
 Vegetable ghee (s)
 - (c) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \xrightarrow{\text{Pt}(\text{s})} 2\text{SO}_3(\text{g})$
 - (d) $\text{C}_{12}\text{H}_{22}\text{O}_{11}(\text{aq}) + \text{H}_2\text{O}(\ell) \xrightarrow{\text{H}_2\text{SO}_4(\text{aq})}$
 $\text{C}_6\text{H}_{12}\text{O}_6 + \text{C}_6\text{H}_{12}\text{O}_6$
- (1) a and b
 - (2) b and c
 - (3) c and a
 - (4) a, b and d

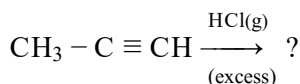
55. Which of the following alkenes on hydration gives a tertiary alcohol

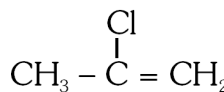
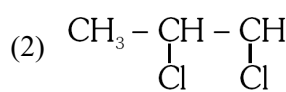
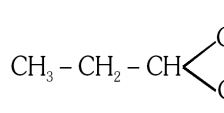
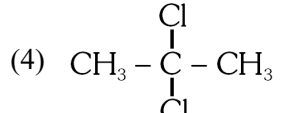
- (1) 2-Butene
- (2) Isobutylene
- (3) Ethene
- (4) α -Butylene

56. 

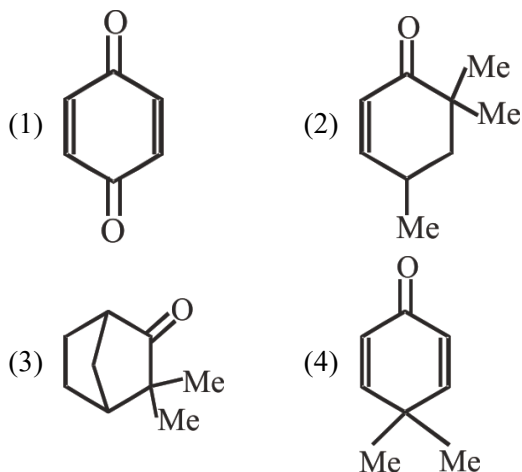
- (1) 
- (2) 
- (3) 
- (4) 

57. What is the main product of this reaction?

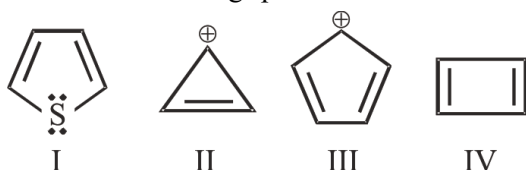


- (1) 
- (2) 
- (3) 
- (4) 

58. Tautomerism is shown by the following compounds :-

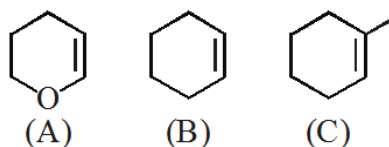


59. Which of the following species are anti-aromatic ?

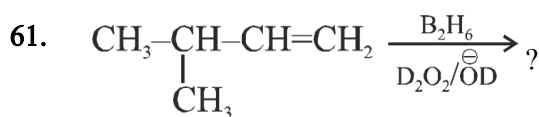


- (1) III and IV (2) I and III
(3) I and II (4) I and IV

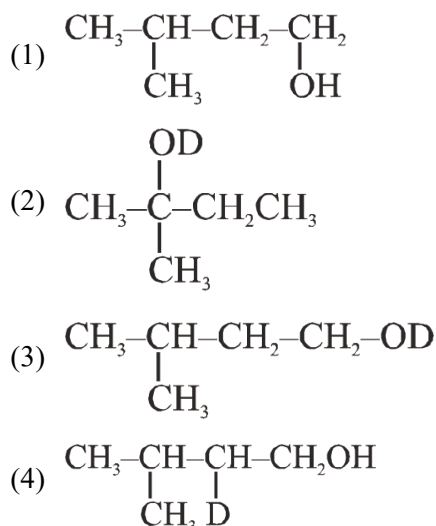
60. Rate of reaction towards EAR is :-



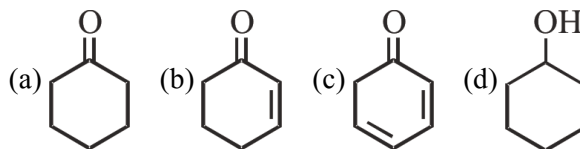
- (1) A > B > C (2) B > A > C
(3) C > A > B (4) A > C > B



Identify the product :-

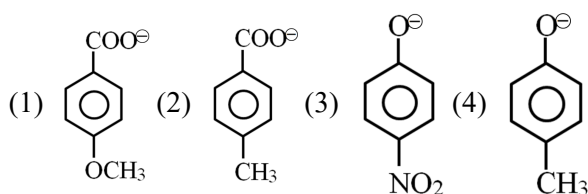


62. Compare bond length of C-O :-

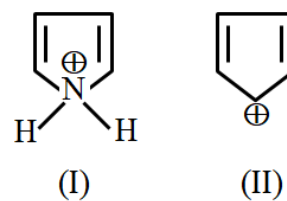


- (1) a > b > c > d
(2) d > b > c > a
(3) d > c > b > a
(4) d > a > c > b

63. Which of the following is most stable anion :-

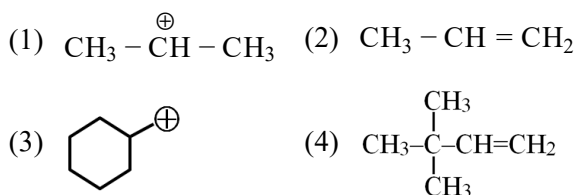


64. Which statement is correct for following compound :-

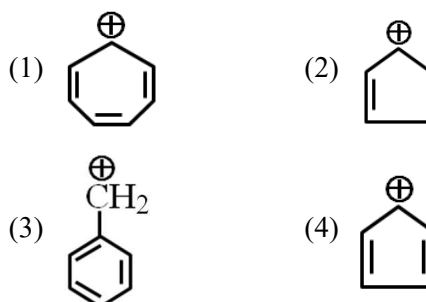


- (1) I & II are aromatic
(2) I & II are antiaromatic
(3) I is aromatic and II is antiaromatic
(4) I is non-aromatic and II is antiaromatic

65. Hyper conjugation is absent in which of the following compound :-



66. Most stable carbocation is :-



67. **Assertion** :- If in a zero order reaction, the concentration of the reactant is doubled, the half life period is also doubled.

Reason :- For a zero order reaction, the rate of reaction is independent of initial concentration.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

68. **Assertion** :- Salt like KCl, KNO₃ i.e. inert electrolytes are used in salt Bridge.

Reason :- An inert electrolyte can directly be filled in U-tube.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

69. For the non-stoichiometric reaction
 $2A + B \rightarrow C + D$, the following kinetic data were obtained in three separate experiments, all at 298 K.

Initial concentration (A)	Initial concentration (B)	Initial rate of formation of C ($\text{mol L}^{-1}\text{s}^{-1}$)
0.1 M	0.1 M	1.2×10^{-3}
0.1 M	0.2 M	1.2×10^{-3}
0.2 M	0.1 M	2.4×10^{-3}

The rate law for the formation of C is :

- (1) $\frac{dC}{dt} = k[A][B]$
- (2) $\frac{dC}{dt} = k[A]^2[B]$
- (3) $\frac{dC}{dt} = k[A][B]^2$
- (4) $\frac{dC}{dt} = k[A]$

70. Mechanism of a hypothetical reaction :



- (i) $X_2 \rightarrow X + X$ (fast)
- (ii) $X + Y_2 \rightleftharpoons XY + Y$ (slow)
- (iii) $X + Y \rightarrow XY$ (fast)

The overall order of the reaction will be :

- (1) 2
- (2) 0
- (3) 1.5
- (4) 1

71. The rate constant for a zero order reaction is $2 \times 10^{-2} \text{ mol L}^{-1} \text{ sec}^{-1}$. If the concentration of the reactant after 25 sec is 0.5 M, the initial concentration must have been :

- (1) 0.5 M
- (2) 1.25 M
- (3) 12.5 M
- (4) 1.0 M

72. Rate of a general reaction $A + B \rightarrow \text{products}$ can be expressed as follows on the basis of collision theory.

$$\text{Rate} = Z_{AB}e^{-E_a/RT}$$

Which of the following statements is not correct for the above expression ?

- (1) Z is collision frequency and is equal to number of collisions per second per unit volume of the reaction mixture.
- (2) $e^{-E_a/RT}$ is the fraction of molecules with kinetic energy equal to or greater than E_a .
- (3) E_a is activation energy of the reaction.
- (4) All the molecules which collide with one other are effective collisions.

73. The activation energy of the reaction,
 $A + B \rightarrow C + D + 38 \text{ kcal}$ is 20 kcal. What would be the activation energy of the reaction,
 $C + D \rightarrow A + B$

- (1) 20 kcal
- (2) -20 kcal
- (3) 18 kcal
- (4) 58 kcal

74. At 300 K the half-life of a sample of a gaseous compound initially at 1 atm is 100 sec. When the pressure is 0.5 atm the half-life is 50 sec. The order of reaction is :

- (1) 0
- (2) 1
- (3) 2
- (4) 3

75. For a reaction of order n , the unit of the rate constant is :

- (1) $\text{mol}^{1-n} \text{L}^{1-n} \text{s}$
- (2) $\text{mol}^{1-n} \text{L}^{2n} \text{s}^{-1}$
- (3) $\text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$
- (4) $\text{mol}^{1-n} \text{L}^{1-n} \text{s}^{-1}$

76. Which of the following reaction is elementary ?

- (1) $2A + B \rightarrow 3C, r = k[A]^1[B]^2$
- (2) $2A + 2B \rightarrow C, r = k[A]^2[B]^3$
- (3) $A + B \rightarrow C, r = k[A]^1[B]^1$
- (4) $2A \rightarrow B, r = k[A]^0$

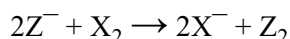
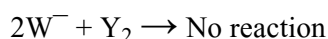
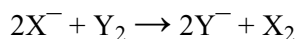
77. In a reaction between P & Q if concentration of P alone is doubled rate increases 4 times and on doubling the concentration of both P & Q the rate becomes 8 times. The rate law of the reaction is :-

- (1) $r = k[P]^2[Q]$
- (2) $r = k[P][Q]^2$
- (3) $r = k[P][Q]$
- (4) $r = k[P][Q]^3$

78. Correct expression of the rate constant for zero order reaction is :

- (1) $k = \frac{[A]_0}{2}t$
- (2) $k = \frac{1}{t} \{ [A]_0 - [A] \}$
- (3) $k = \frac{1}{t} \{ [A] - [A]_0 \}$
- (4) $k = \frac{2.303}{t} \log \{ [A]_0 - [A] \}$

79. The following facts are available



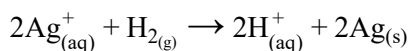
Which of the following statements is correct?

- (1) $E_{W^-|W_2}^\circ > E_{Y^-|Y_2}^\circ > E_{X^-|X_2}^\circ > E_{Z^-|Z_2}^\circ$
- (2) $E_{W^-|W_2}^\circ > E_{Y^-|Y_2}^\circ < E_{X^-|X_2}^\circ < E_{Z^-|Z_2}^\circ$
- (3) $E_{W^-|W_2}^\circ < E_{Y^-|Y_2}^\circ > E_{X^-|X_2}^\circ > E_{Z^-|Z_2}^\circ$
- (4) $E_{W^-|W_2}^\circ < E_{Y^-|Y_2}^\circ < E_{X^-|X_2}^\circ < E_{Z^-|Z_2}^\circ$

80. Solution of KNO_3 is used in salt Bridge because :-

- (1) Velocity of K^+ is greater than that of NO_3^-
- (2) Velocity of NO_3^- is greater than that of K^+
- (3) Velocity of both K^+ and NO_3^- are nearly same
- (4) KNO_3 is highly soluble in water

81. The cell reaction



is best represented by :-

- (1) $\text{Ag}_{(\text{s})}|\text{Ag}_{(\text{aq})}^+||\text{H}_{(\text{aq})}^+|\text{H}_{2(\text{g})}|\text{Pt}_{(\text{s})}$
- (2) $\text{Pt}_{(\text{s})}|\text{H}_{2(\text{g})}|\text{H}_{(\text{aq})}^+||\text{Ag}_{(\text{aq})}^+|\text{Ag}_{(\text{s})}$
- (3) $\text{Ag}_{(\text{s})}|\text{Ag}_{(\text{aq})}^+||\text{H}_{2(\text{g})}|\text{H}_{(\text{aq})}^+|\text{Pt}_{(\text{s})}$
- (4) $\text{Pt}_{(\text{s})}|\text{H}_{(\text{aq})}^+|\text{H}_{2(\text{g})}||\text{Ag}_{(\text{s})}|\text{Ag}_{(\text{aq})}^+$

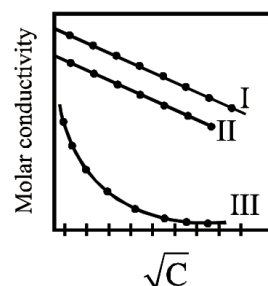
82. Equivalent conductivity of $\text{Fe}_2(\text{SO}_4)_3$ is related to molar conductivity by the expression :-

- (1) $\Lambda_{\text{eq}} = \Lambda_{\text{m}}$
- (2) $\Lambda_{\text{eq}} = \Lambda_{\text{m}}/3$
- (3) $\Lambda_{\text{eq}} = 3\Lambda_{\text{m}}$
- (4) $\Lambda_{\text{eq}} = \Lambda_{\text{m}}/6$

83. The equivalent conductivity at infinite dilution (Λ^∞) for sodium chloride, HCl and sodium acetate in water at 298K are 128.0, 425.0 and 91.0 $\text{Scm}^2\text{eq}^{-1}$ respectively. A solution of acetic acid shows an equivalent conductivity of 38.8 $\text{Scm}^2\text{eq}^{-1}$. What is the percent dissociation of acetic acid :-

- (1) 5.0
- (2) 10.0
- (3) 20.0
- (4) 30.0

84. A graph was plotted between molar conductivity of various electrolytes (NaCl, HCl and NH_4OH) and $\sqrt{C}$ (in mol L^{-1}). Correct setting :-



- (1) I(NaCl), II(HCl), III(NH_4OH)
- (2) I(HCl), II(NaCl), III(NH_4OH)
- (3) I(NH_4OH), II(NaCl), III(HCl)
- (4) I(NH_4OH), II(HCl), III(NaCl)

85. Specific conductance is :-

- (1) reciprocal of resistance
- (2) reciprocal of conductance
- (3) conductance of 1 ml solution
- (4) all of above

SECTION-B (CHEMISTRY)

86. 100 ml of a colloidal solution is completely precipitated by addition of 5ml of 1M NaCl solution. Calculate the coagulation value of NaCl

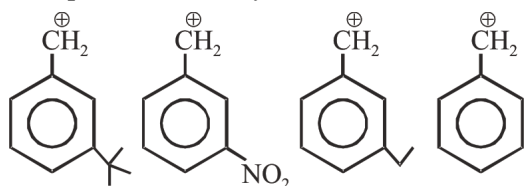
- (1) 100
- (2) 5
- (3) 50
- (4) 120

	Column-I		Column-II
			TG: @Chalnaayaaar
(a)	Precipitation of colloidal sol	(p)	Peptization
(b)	Purification of colloid	(q)	Coagulation
(c)	Gold number	(r)	Desorption
(d)	Formation of colloidal sol	(s)	Dialysis
		(t)	Measure of protective power

Which of the following option show correct matches :

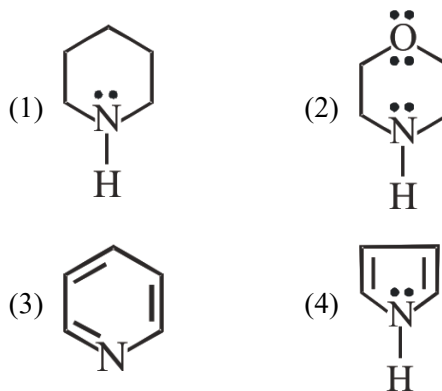
- (1) (a-q), (b-s), (c-r), (d-p)
- (2) (a-s), (b-q), (c-p), (d-r)
- (3) (a-r), (b-s), (c-q), (d-t)
- (4) (a-q), (b-s), (c-t), (d-p)

88. Compare the stability of carbocation :-



- (1) 2 > 4 > 3 > 1
- (2) 1 > 3 > 4 > 2
- (3) 3 > 1 > 4 > 2
- (4) 4 > 3 > 2 > 1

89. Least basic among the following is :-



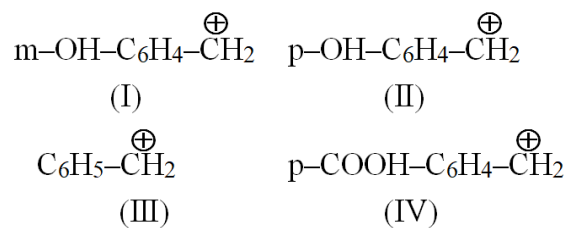
90. Primary alcohol is formed as a major product in :-

- (1) $\text{CH}_3-\text{CH}=\text{CH}_2 \xrightarrow{\text{H}^+/\text{H}_2\text{O}}$
- (2) $\text{CH}_3-\text{CH}=\text{CH}_2 \xrightarrow[(2)\text{H}_2\text{O}_2/\bar{\text{O}}\text{H}]{(1)\text{BH}_3/\text{THF}}$
- (3) $\text{CH}_3-\text{CH}=\text{CH}_2 \xrightarrow[(2)\text{NaBH}_4]{(1)(\text{CH}_3\text{COO})_2\text{Hg}/\text{H}_2\text{O}}$
- (4) None of these

91. If 'X' substituent present at para position of benzoic acid decreases its $\text{P}k_a$, then "X" is :-

- (1) $-\text{OCH}_3$
- (2) $-\text{CH}_3$
- (3) $-\text{OH}$
- (4) $-\text{CN}$

92. The decreasing order of stability of following carbocation is :-



- (1) I > II > III > IV
- (2) II > III > IV > I
- (3) III > II > I > IV
- (4) II > III > I > IV

93. Rate constant for a chemical reaction takes place at 500K is expressed as $k = A.e^{-1000}$.

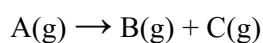
The activation energy of the reaction is :

- (1) 100 cal/mol
- (2) 1000 kcal/mol
- (3) 10^4 kcal/mol
- (4) 10^6 kcal/mol

94. The value of rate constant of a pseudo first order reaction

- (1) Depends on the concentration of reactants present in small amount.
- (2) Depends on the concentration of reactants present in excess.
- (3) Is independent of the concentration of reactants.
- (4) Depends only on temperature.

95. Consider a first order gas phase decomposition reaction given below :



The initial pressure of the system before decomposition of A was p_i . After lapse of time 't', total pressure of the system increased by x units and become ' p_t '. The rate constant k for the reaction is given as

- (1) $k = \frac{2.303}{t} \log \frac{p_i}{2p_i - x}$
- (2) $k = \frac{2.303}{t} \log \frac{p_i}{2p_i - p_t}$
- (3) $k = \frac{2.303}{t} \log \frac{p_i}{2p_i + p_t}$
- (4) $k = \frac{2.303}{t} \log \frac{p_i}{p_i + x}$

96. Consider the Arrhenius equation given below and mark the correct option :

$$k = Ae^{-E_a/RT}$$

- (1) Rate constant increases exponentially with increasing activation energy and decreasing temperature.
- (2) Rate constant decreases exponentially with decreasing activation energy and decreasing temperature.
- (3) Rate constant increases exponentially with decreasing activation energy and decreasing temperature.
- (4) Rate constant increases exponentially with decreasing activation energy and increasing temperature.

97. Ionisation constant of a weak acid (HA) in terms of Λ_m^∞ and Λ_m is :-

- (1) $K_a = \frac{C\Lambda_m^\infty}{(\Lambda_m - \Lambda^\infty)}$
- (2) $K_a = \frac{C\Lambda_m^2}{\Lambda_m^\infty(\Lambda_m^\infty - \Lambda_m)}$
- (3) $K_a = \frac{C(\Lambda_m^\infty)^2}{\Lambda_m^\infty(\Lambda_m - \Lambda_m^\infty)}$

(4) None of these

98. If specific conductance and conductance of a solution is same, then the cell constant is equal to :-

- (1) 1
- (2) 0
- (3) 10
- (4) 100

99. What will be the molar conductance " Λ_m ", if resistivity is 'x' for 0.1 N H_2SO_4 solution?

- (1) $\Lambda_m = \frac{x \times 1000}{0.1}$
- (2) $\Lambda_m = \frac{2 \times 1000}{x \times 0.1}$
- (3) $\Lambda_m = \frac{x \times 1000}{0.5}$
- (4) $\Lambda_m = \frac{0.5}{1000x}$

100. Metals X, Y, Z have standard oxidation potentials $-0.65V$, $-1.40V$ and $+0.36V$ respectively then :-

- (1) Z can oxidise X but not Y.
- (2) Y can oxidise X but not Z.
- (3) Z can reduce $X^\oplus$ and $Y^\oplus$ both.
- (4) $X^\oplus$ can reduce Z but not Y.

Topic : HUMAN HEALTH AND DISEASE, ENVIRONMENTAL ISSUES AND LIFE CYCLE OF ANGIOSPERM (UPTO POLLINATION), ORIGIN AND EVOLUTION (UPTO DARWIN THEORY) [BACK UNIT] : HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH

TG: @Chalnaayaaar

SECTION-A (BIOLOGY-I)

101. Which of the following statements are correct?
 (A) Reproductive and Child Health Care (RCH) programme was initiated in 1951.
 (B) A rapid decline in death rate, maternal mortality rate (MMR) and infant mortality rate (IMR) are also probable reasons for increase in population.
 (C) Natural methods of contraception work on the principle of avoiding chances of ovum and sperms meeting.
 (D) If duration of pregnancy is between 12 to 24 months then opinion of 2 registered medical practitioners is needed for MTP.
- (1) A, B, C & D (2) A, C & D
 (3) B, C & D (4) C and D
102. **Statement-I** : Progestasert is hormone releasing IUD.
Statement-II : IUD decrease phagocytosis of sperm within uterus.
- (1) Both statements are correct
 (2) Both statements are incorrect
 (3) Statement-I is correct, statement-II is incorrect
 (4) Statement-I is Incorrect, statement-II is correct
103. **Assertion** :- Artificial insemination (AI) is a technique of in vivo fertilisation.
Reason :- In (AI) technique, semen collected either from husband or healthy donar and artificially introduced in vagina or uterus.
- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) Assertion is True but the Reason is False.
 (4) Both Assertion & Reason are False.

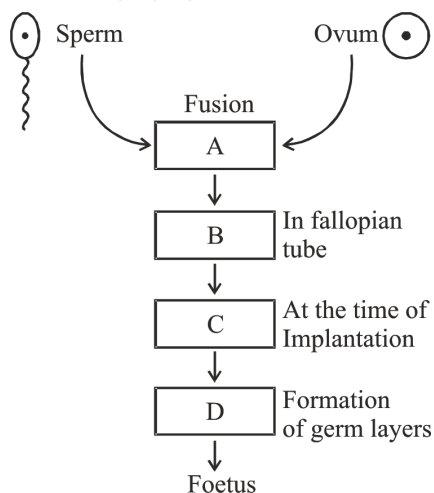
104. Which technologies in the list given below are showing in-vivo fertilisation?
 GIFT, ZIFT, ICSI, IUT, AI, IUI, IVF
- (1) GIFT, ZIFT, ICSI, IUT
 (2) GIFT, AI, IUI
 (3) IUT, AI, IUI, IVF
 (4) ZIFT, ICSI, IUT
105. The following statements are made regarding MTP. Choose the correct option given below
 (i) MTPs are generally advised during the first trimester
 (ii) MTPs are used as contraceptive method
 (iii) MTPs are always surgical
 (iv) MTPs require the assistance of qualified medical personnel
- (1) (ii) and (iii) (2) (i) and (iii)
 (3) (i) and (iv) (4) (i) and (ii)

106. Identify the correct match

(1)	Proliferative phase	-	Formation of corpus luteum
(2)	Secretory phase	-	Development of tertiary follicle
(3)	Bleeding phase	-	Shedding of stratum functionalis
(4)	Ovulatory phase	-	Release of primary oocyte

107. Read the following statements :
- (I) Oxytocin is a milk let down hormone.
 (II) Relaxin causes strong contraction in uterine placenta during parturition.
 (III) Oxytocin cause strong uterine contraction during parturition.
 (IV) Presence of hCG in maternal blood or urine is indicator of pregnancy.
- How many statements are correct :
- (1) One (2) Three (3) Four (4) Two

108. Study the cycle shown below and select the option which gives correct words for all four blanks A, B, C, D :-

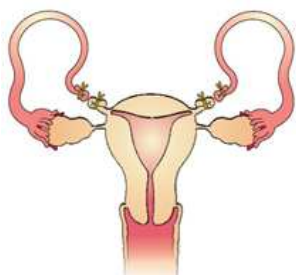


	A	B	C	D
(1)	Foetus	Zygote	Morula	Gastrula
(2)	Zygote	Morula	Blastocyst	Gastrula
(3)	Embryo	Blastocyst	Gastrula	Cleavage
(4)	Zygote	Morula	Gastrula	Cleavage

109. How many devices are include in Intra uterine devices. Nim-76, Today, Lippes loop, Multiload 375, Cu-T, LNG-20, Condoms, Diaphragms, Cervical Cap, Vaults

(1) Four (2) Five (3) Eight (4) Six

110. Select the incorrect statement for given figure:



- (1) This is a permanent method of contraception in females
 (2) Menstrual flow isn't affected after using this method of contraception
 (3) Maturation of Graafian follicles permanently ceases after applying this contraceptive method
 (4) Reversibility of this technique is very poor

111. The chances of fertilization are very high during which days of menstruation cycle ?

(1) First 5 days
 (2) Last 7 days
 (3) 10th to 17th day
 (4) 21st to 25th day

112. Inability of male partner to inseminate the female or due to very low sperm count in the ejaculate, could by corrected by using :-

(1) Gamete Intra Fallopian Transfer (GIFT)
 (2) Artificial Insemination (AI)
 (3) Zygote Intra Fallopian Transfer (ZIFT)
 (4) Intra Uterine Transfer (IUT)

113. Amniocentesis is employed for determine :-

(1) Cardiac abnormality of embryo
 (2) Hereditary abnormality in embryo
 (3) Errors in amino acid metabolism in embryo
 (4) Skin abnormalities

114. In which method of contraception, progesterone can not be used :-

(1) Emergency contraception
 (2) Oral contraceptive pills
 (3) IUD
 (4) Barrier method of female

115. Consider the following four statements (I-IV) reagarding kidney transplant and select the two correct ones out of these.

I. Even if a kidney transplant is proper the recipient may need to take immunosuppressants for a long time
 II. The cell-mediated immune response is responsible for the graft rejection.
 III. The B-lymphocytes are responsible for rejection of the graft.
 IV. The acceptance or rejection of a kidney transplant depends on specific interferons.

(1) II and III (2) III and IV
 (3) I and III (4) I and II

116. When you have gone to a new place and suddenly you started sneezing, wheezing for no explained reason, and when you went away, your symptoms disappeared. The above mentioned reaction could be because of :-

- (1) Viral infection
- (2) Bacterial infection
- (3) Allergy to pollen and mites
- (4) Auto-immunity

117. Choose the correct match :-

- (1) First line defence - fever
- (2) Second line defence - CMI
- (3) Third line defence - AMI
- (4) Third line defence - pH of vagina

118. Match the following column A and B ?

	A		B
1.	IgA	a.	Dimeric
2.	IgG	b.	Pentameric
3.	IgE	c.	Allergic
4.	IgM	d.	Universal

- (1) 1-a, 2-b, 3-c, 4-d
- (2) 1-a, 2-d, 3-b, 4-c
- (3) 1-a, 2-d, 3-c, 4-b
- (4) 1-d, 2-a, 3-c, 4-b

119. Where immature lymphocytes differentiate into antigen sensitive lymphocytes ?

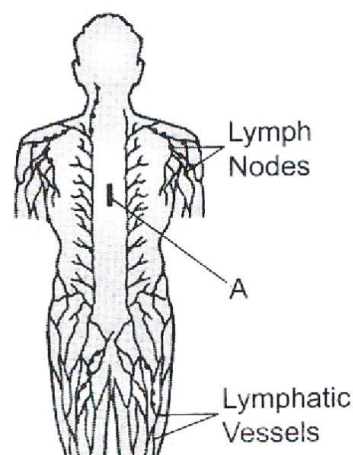
- (1) Spleen
- (2) Lymph Nodes
- (3) Bone Marrow
- (4) 1 & 3 both

120. Match the column-I with column-II and choose the correct options :-

Column-I (Disease)		Column-II (Antibodies formed against)	
(A)	Hashimoto's Disease	(i)	Castle intrinsic factor
(B)	Myasthenia gravis	(ii)	Thyroid gland
(C)	Pernicious Anemia	(iii)	β -cells of pancreas
(D)	Insulin dependent diabetes mellitus (I.D.D.M.)	(iv)	Acetylcholine receptors

- (1) A-iii, B-ii, C-iv, D-i
- (2) A-iv, B-iii, C-i, D-ii
- (3) A-ii, B-iv, C-i, D-iii
- (4) A-ii, B-i, C-iv, D-iii

121. Identify (A) from diagram here and choose the correct option :



- (1) Thymus-Endocrine gland
- (2) Thyroid-Endocrine gland
- (3) Thymus-Primary lymphoid organ
- (4) Both (1) and (3)

122. Which of the following is a main lymphoid organ.

- (1) Thymus gland
- (2) Bone marrow
- (3) Tonsils
- (4) Spleen

- 123.** MALT is located in -
 (1) Respiratory Tract
 (2) Digestive Tract
 (3) Urogenital Tract
 (4) All of these
- 124.** Which of the following drug reduce the symptoms of allergy :
 (1) Anti Histamines
 (2) Adrenalin
 (3) Steroids
 (4) All of these
- 125.** In which of the following auto immune disorder antibodies form against synovial membrane of joints.
 (1) Vitiligo
 (2) Rheumatoid arthritis
 (3) Psoriasis
 (4) Coeliac disease
- 126.** Select the incorrect option for primary lymphoid organs -
 (1) Provide micro environment for development and maturation of lymphocytes.
 (2) Include an endocrine gland located beneath the breast bone
 (3) Immature lymphocyte differentiate into antigen sensitive lymphocyte
 (4) Store lymphocyte and phagocytes.
- 127.** Enlarged end of penis is called
 (1) Glans penis
 (2) Prepuce
 (3) Fore skin
 (4) Corpora cavernosa
- 128.** Which of the following structure is not located in pelvic region :
 (1) External genitalia (2) Ovaries
 (3) Mammary gland (4) Vagina
- 129.** Select the correct option for endometrium.
 (1) Innermost layer of uterus
 (2) Glandular layer
 (3) Lines the uterine cavity
 (4) All of these
- 130.** Select the incorrect option for Bronchial Asthma.
 (1) Allergic disorder
 (2) Spasm of Bronchi & Bronchiole present
 (3) Affect upper respiratory tract
 (4) Wheezing sound is produced
- 131.** Which of the following provide artifical passive immunity .
 (1) Anti diptherial serum
 (2) Vaccine
 (3) IgG from mother
 (4) IgA from mother
- 132.** The Big bang theory attempts to explain us the origin of:
 (1) Life
 (2) Universe
 (3) Human
 (4) All of the above
- 133.** Famous swan neck experiment was done by Louis Pasteur. Which theory was proved by this experiment :-
 (1) Abiogenesis
 (2) Special creation theory
 (3) Biogenesis
 (4) Cosmic panspermia theory

134. Life was created by some supernatural power this statement supports to :-

- (1) Oparin-Haldane theory
- (2) Cosmic panspermia
- (3) Special creation theory
- (4) Biogenesis

135. According to life came from outer space. Fill in the blanks with appropriate option:

- (1) Special creation theory
- (2) Cosmic panspermia theory
- (3) Spontaneous generation theory
- (4) Biogenesis theory

SECTION-B (BIOLOGY-I)

136. **Assertion (A)** : Saheli is non steroidal oral-contraceptive pill for female.

Reason (R) : It has to be taken within 72 hours of unprotected coitus.

- (1) Both Assertion (A) and Reason (R) are correct and (R) is correct explanation of (A).
- (2) Both Assertion (A) and Reason (R) are correct and (R) is not correct explanation of (A).
- (3) (A) is correct but (R) is incorrect.
- (4) (A) is incorrect but (R) incorrect.

137. Which of following is correct regarding the effect of vasectomy:-

- (1) Infertility occurs in male.
- (2) Semen doesn't released out.
- (3) Libido is lost.
- (4) Penis is removed.

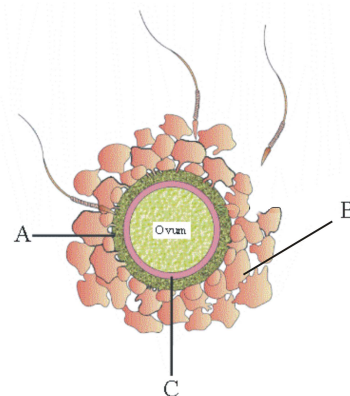
138. The correct match w.r.t. the source gland and its respective hormone as well as its function is represented by which option?

	Source Gland	Hormone	Function
(1)	Anterior pituitary	Vasopressin	Increases blood pressure
(2)	Posterior pituitary	Luteinizing hormone	Initiate the ejection of milk
(3)	Corpus albicans	FSH	Supports pregnancy
(4)	Hypothalamus	Somatocrinin / GHRH	Stimulates the release growth hormone

139. The correct travelling pathway of ovum in human:

- (1) Ovary → coelom → infundibulum → ampulla → isthmus → uterus
- (2) Ovary → infundibulum → ampulla → isthmus → coelom → uterus
- (3) Ovary → coelom → ampulla → infundibulum → isthmus → uterus
- (4) Ovary → coelom → isthmus → ampulla → infundibulum → uterus

140. The given below diagram represents "ovum" surrounded by few sperms. Identify A,B,C :-



	A	B	C
(1)	Zona pellucida	Perivitteline space	Corona radiata
(2)	Zona pellucida	Corona radiata	Perivitteline space
(3)	Corona radiata	Perivitteline space	Zona pellucida
(4)	Corona radiata	Zona pellucida	Perivitteline space

141. Which of the following pair is correct :-

		Fertilisation	ART
(1)	a	In vitro	ZIFT, IUT, ICSI
	b	In vivo	GIFT, AI
(2)	a	In vivo	ZIFT, IUT, ICSI
	b	In vitro	GIFT, AI
(3)	a	In vitro	GIFT, IUT, ICSI
	b	In vivo	IUT, AI
(4)	a	In vivo	ZIFT, IUT, ICSI
	b	In vitro	IUT, AI

142. Which of the following is barrier contraceptive of females, that can prevent against sexually transmitted infections ?

- (1) Cervical cap
- (2) Vault
- (3) Diaphragm
- (4) Female condom

143. Which of the following is/are hormone releasing IUDs ?

- (1) Progestasert
- (2) Levonorgestrel
- (3) Both (1) & (2)
- (4) Lippe's loop

144. Identify the correct statements ?

- (A) Saliva in the mouth considered in physiological barrier.
- (B) Interferones are virus specific.
- (C) Our immune system feeds the memory of first encounter with pathogen.
- (D) T-cells do not secrete antibodies but help B-cell to produce them.
- (E) Antibodies are found in blood that's why their response is called humoral immune response.

- (1) Five
- (2) Four
- (3) Three
- (4) One

145. Gestation period is maximum in

- (1) Cat
- (2) Dog
- (3) Human
- (4) Elephant

146. Select the incorrect option for implant :

- (1) Effective period much longer than pills.
- (2) Mode of action is like CuT.
- (3) Inserted under the skin.
- (4) Progesterone alone or progesterone estrogen combination used by female as implants.

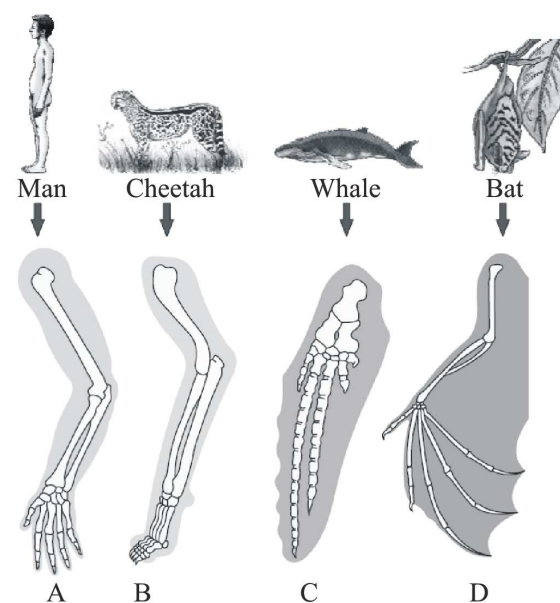
147. In Tubectomy fallopian tube is removed or tied up through a small incision in the :

- (1) Abdomen
- (2) Thorax
- (3) Kidney
- (4) Both 1 & 3

148. In 2011 world population was

- (1) 2 billion
- (2) 350 billion
- (3) 7.2 billion
- (4) 6 billion

149. Which of the following is correct for the structures A, B, C and D shown in the diagram ?



- (1) A, B, C and D are analogous structures
- (2) A, B, C and D shows divergent evolution
- (3) A, B, C and D show convergent evolution
- (4) A, B, C and D are vestigial structures

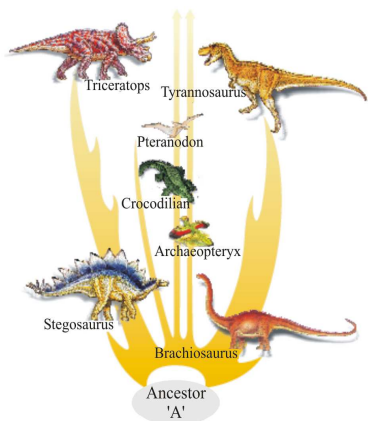
150. Which of the following is a correct set of vestigial organs :-

- (1) Coccyx, vermiform appendix, ear pinna
- (2) Nails, plica semilunaris, coccyx, pinna
- (3) Ear muscle, patella, caecum
- (4) Pinna muscle, body hair, canine, wisdom tooth

Topic : HUMAN HEALTH AND DISEASE, ENVIRONMENTAL ISSUES AND LIFE CYCLE OF ANGIOSPERM (UPTO POLLINATION), ORIGIN AND EVOLUTION (UPTO DARWIN THEORY) [BACK UNIT] : HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH

SECTION-A (BIOLOGY-II)

151. Identify the ancestor-A in the family tree of dinosaur in given figure :-



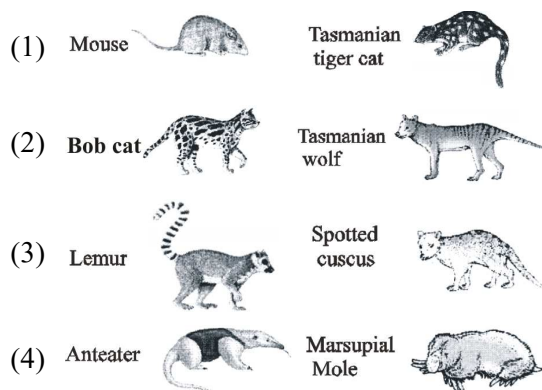
- (1) Synapsids (2) Thecodont
(3) Therapsids (4) Pelycosaurus
152. Which of the following provide most evident proof of evolution ?
- (1) Morphology (2) Embryo
(3) Vestigial organs (4) Fossils
153. What is the correct sequence of origin of Mammals ?
- (1) Synapsids → Pelycosaurs → Therapsids → Mammals
(2) Sauropsids → Pelycosaurs → Therapsids → Mammals
(3) Sauropsids → Synapsids → Thecodont → Mammals
(4) Synapsids → Sauropsids → Therapsids → Mammals
154. About 200 mya, some reptiles moved back into water and evolved into :-
- (1) Mammals
(2) Penguin
(3) Fish-like reptiles – *Ichthyosaurs*
(4) Amphibians

155. Match the following columns and select the correct option accordingly :-

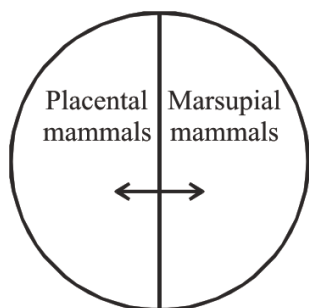
Column-I	Column-II
(A) Golden Age of fishes	(i) Coenozoic era
(B) Golden Age of dinosaurs	(ii) Carboniferous period
(C) Golden age of amphibians	(iii) Devonian period
(D) Golden Age of birds and mammals.	(iv) Jurassic period

- (1) A – ii, B – iv, C – iii, D – i
(2) A – iii, B – iv, C – i, D – ii
(3) A – iii, B – iv, C – ii, D – i
(4) A – ii, B – iv, C – i, D – iii
156. Invertebrates evolved and become active about :-
- (1) 50 mya
(2) 500 mya
(3) 5000 mya
(4) 5 mya
157. Amphibians supposed to be evolved from the most primitive fish that could move on land and water, this was :-
- (1) Jawless fish
(2) Coelacanth
(3) Chimera
(4) Ichthyosaurs
158. Which of the following are not analogous organs :
- (1) Fins of fishes and flippers of whales
(2) Stings of honey bee and sting of scorpion
(3) Thorn of *Bougainvillea* and tendril of *Cucurbita*
(4) Wings of insects and wings of birds

159. In the given figure which one of the following represent the correct example of convergent evolution ?



160. Given figure shows :-



- (1) Adaptive radiations of marsupials
- (2) Adaptive radiations of placental mammals
- (3) Convergent evolution between marsupials and placental mammals
- (4) Divergent evolution between marsupials and placental mammals

161. Green Mufflar is :-

- (1) Green plant are planted for checking noise pollution
- (2) Green house effect
- (3) Green revolution
- (4) Green plant are planted for checking all air pollutants

162. O_3 + PAN + Nitrogen oxides, form :-

- (1) London/Sulphur smog
- (2) Los Angeles smog/Photochemical smog
- (3) FOAM
- (4) Minamata

163. Which one of the following can remove over 99 percent particulate matter present in the exhaust from a thermal power plant ?

- (1) Scrubber
- (2) Catalytic converters
- (3) Electrostatic precipitator
- (4) Green muffler

164. FOAM are concerned with :-

- (1) Eutrophication
- (2) Integrated waste water treatment
- (3) Biomagnification
- (4) Remedy for solid waste

165. Three Mile Island and Chernobyl incident are related to :-

- (1) Soil pollution
- (2) Water pollution
- (3) Air pollution
- (4) Radioactive pollution

166. Most problematic weed of the world present in most of the pond, lake and streams is :-

- (1) *Parthenium*
- (2) *Nile perch*
- (3) *Lantana*
- (4) *Eichhornia crassipes*

167. Water pollution Act was passed by the Government of India in :-

- | | |
|----------|----------|
| (1) 1974 | (2) 1984 |
| (3) 1972 | (4) 1992 |

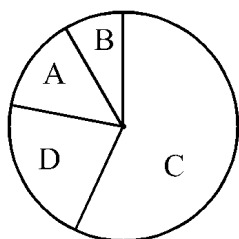
168. "Good Ozone" is found in the :-

- (1) Stratosphere
- (2) Ionosphere
- (3) Mesosphere
- (4) Troposphere

169. Global agreement in specific control strategies to reduce the release of ozone depleting substances, was adopted by :-

- (1) The Koyoto Protocol
- (2) The Vienna Convention
- (3) Rio de Janeiro Conference
- (4) The Montreal Protocol

170. Relative contribution of various greenhouse gases to total global warming is given in circle. Indentify correct option.



- (i) C is-major product produce by burning of biomass
- (ii) D is-main responsible gas for ozone depletion
- (iii) A is-most abundant natural hydrocarbon
- (iv) B is-Release from fertilizer industries

- (1) i, ii, iii, iv
- (2) i, iv
- (3) iii, iv
- (4) i, ii

171. Match the items in column I and column II and choose the correct option :-

Column I		Column II	
A.	UV	(i)	Biomagnification
B.	Biodegradable organic matter	(ii)	Eutrophication
C.	DDT	(iii)	Snow blindness
D.	Phosphates	(iv)	BOD

The correct match is :-

- (1) A-ii, B-i, C-iv, D-iii
- (2) A-iii, B-ii, C-iv, D-i
- (3) A-iii, B-iv, C-i, D-ii
- (4) A-iii, B-i, C-iv, D-i

172. Choose the correct option :-

- (1) Algal bloom – Biomagnification
- (2) Minamata disease – cadmium (Cd) poisoning
- (3) Arcata marsh – Heavy metal pollution
- (4) Good ozone – Troposphere

173. According to Central Pollution Control Board (CPCB), which particulate size in diameter (in micrometers) of the air pollutants is responsible for greatest harm to human health ?

- (1) 25 μm or less
- (2) 5.2 – 2.5 μm
- (3) 2.5 μm or less
- (4) 2.5 μm or more

174. Which one of the following structures diploid in its ploidy level?

- (1) Pollen tetrad
- (2) Embryo sac
- (3) Perisperm
- (4) Antipodal cell

175. Given below are two statements:

- Statement-I** : Cleistogamous flowers produce assured seed-set even in the absence of pollinators.
Statement-II : Cleistogamous flowers are invariably autogamous.

- (1) Both Statement-I and Statement-II are incorrect
- (2) Statement-I is correct but Statement-II is incorrect
- (3) Statement-I is incorrect but Statement-II is correct
- (4) Both Statement-I and Statement-II are correct

176. Which one of the following layer nourishes the developing pollen grains ?

- (1) Endothecium
- (2) Tapetum
- (3) Middle layer
- (4) Epidermis

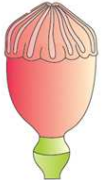
177. The sporogenous tissue (MMCs) undergo meiotic divisions to form microspore tetrad, what would be the ploidy of the cells of the tetrad?

- (1) $2n$
- (2) n
- (3) $4n$
- (4) $3n$

178. The pollen grains represent-

- (1) Male sporophyte
- (2) Male gametophyte
- (3) Female gametophyte
- (4) Female sporophyte

179. Match List-I with List-II :-

	List-I		List-II
1.		(a)	Anatropous ovule
2.		(b)	Pistil of Hibiscus
3.		(c)	Pollen Tetrad
4.		(d)	Pistil of Papaver

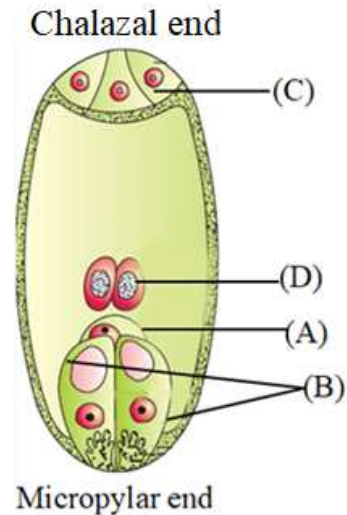
- (1) (1-c); (2-d); (3-b); (4-a)
- (2) (1-c); (2-d); (3-a); (4-b)
- (3) (1-a); (2-b); (3-c); (4-d)
- (4) (1-c); (2-a); (3-b); (4-c)

TG: @Chalnaayaaar

180. The female gametophyte of angiosperm is-

- (1) 7-Celled, 9-Nucleated
- (2) 7-Celled, 8-Nucleated
- (3) 7-Celled, 7-Nucleated
- (4) 8-Celled, 8-Nucleated

181. Find out the structure (A), (B), (C) and (D) in given figure-

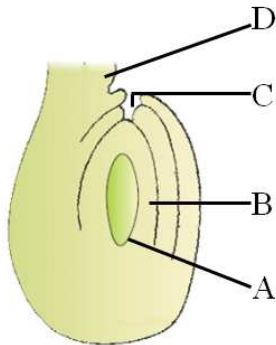


	(A)	(B)	(C)	(D)
(1)	Synergids	Egg	Antipodals	Polar nuclei
(2)	Egg	Antipodal	Synergids	Polar nuclei
(3)	Egg	Synergids	Antipodals	Polar nuclei
(4)	Polar nuclei	Antipodals	Egg	Synergids

182. Which one of the following is correctly matched :-

	List-I		List-II
(1)	Pollen grain	(A)	Diploid
(2)	Embryo sac	(B)	Female gametophyte
(3)	Stamen	(C)	Megasporophyll
(4)	Carpel	(D)	Microsporophyll

183. Find out the structure A, B, C and D in given figure :



	(A)	(B)	(C)	(D)
(1)	Embryo sac	Chalaza	Funicle	Hilum
(2)	Embryo sac	Hilum	Chalaza	Micropyle
(3)	Embryo sac	Nucellus	Funicle	Hilum
(4)	Embryo sac	Nucellus	Micropyle	Funicle

184. Given below are two statements :

Statement-I : Wind and water pollinated flowers are not very colourful and do not produce nector.

Statement-II : Majority of insect-pollinated flowers are large, colourful, fragrant and rich in nector.

Choose the correct answer from the options given below :

- (1) Both Statement-I and Statement-II are false
 - (2) Statement-I is correct but Statement-II is false
 - (3) Statement-I is incorrect but Statement-II is true
 - (4) Both Statement-I and Statement-II are true
185. Which of the following is an example of dioecious plant?

- (1) Castor
- (2) Papaya
- (3) Maize
- (4) Cucumber

SECTION-B (BIOLOGY-II)

186. The presence of notochord and pharyngeal gills slits in embryos of birds and mammals proves :-

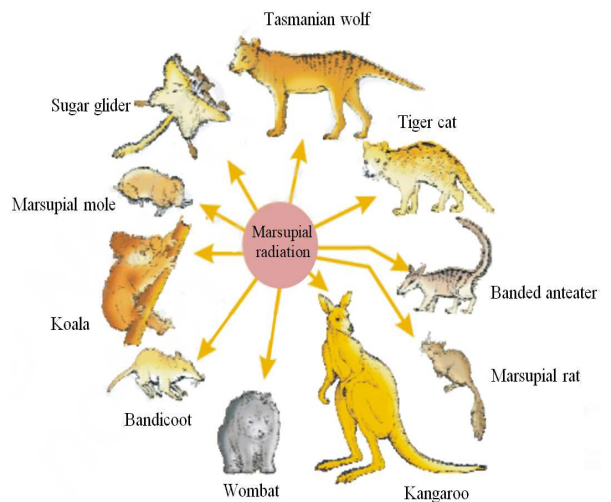
- (1) Convergent evolution
- (2) Mutation
- (3) Ontogeny recapitulates phylogeny
- (4) Co-evolution

187. **Statement-I :** Branching descent and natural selection are two key concept of Darwinian Theory of Evolution.

Statement-II : It is possible that the work of Thomas Malthus on population influenced Darwin.

- (1) Statement-I correct & Statement-II incorrect.
- (2) Statement-I incorrect & Statement-II correct.
- (3) Statement-I correct & Statement-II correct.
- (4) Statement-I incorrect & Statement-II incorrect.

188. How many statements are correct about the diagram given below ?



- (a) Australian marsupials
- (b) Shows Homology
- (c) Example of divergent evolution
- (d) All are examples of adaptive radiation

- (1) one
- (2) two
- (3) three
- (4) four

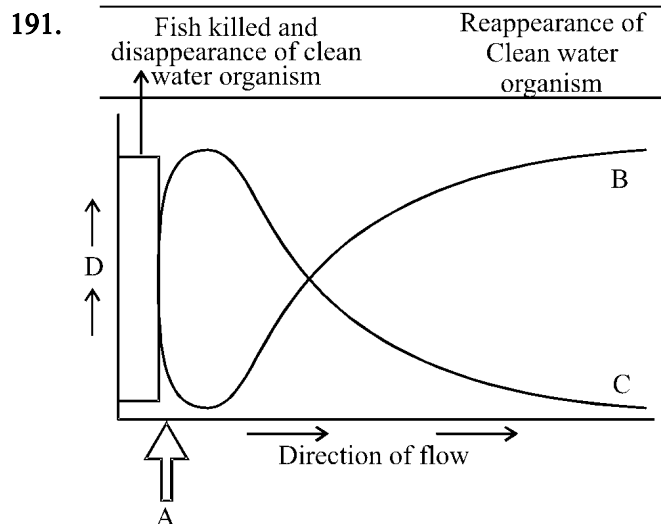
189. Find out the correct sequence of evolution of plants:-

- (1) Tracheophyte ancestors → Rhynia-type plants → Chlorophyte ancestors → Psilophyton
- (2) Chlorophyte ancestors → Tracheophyte ancestors → Rhynia-type plants → Psilophyton
- (3) Psilophyton → Rhynia-type plants → Chlorophyte ancestors → Tracheophyte ancestors
- (4) Psilophyton → Gnetales → Tracheophyte ancestors → Chlorophyte ancestors

190. ___A___ ___B___ and ___C___ of catalytic converter changes unburnt hydrocarbons into CO₂ and water, CO into CO₂ and nitric oxide into nitrogen gas.

A, B and C are :-

- (1) Platinum, Gold and Silver
- (2) Gold, Silver and Rhodium
- (3) Platinum, Palladium and Rhodium
- (4) Mercury, Silver and Palladium



- (1) A–Sewage discharge, B–BOD, C–DO, D–Concentration
- (2) A–Sewage discharge, B–DO, C–BOD, D–Concentration
- (3) A–Concentration, B–BOD, C–DO, D–Sewage discharge
- (4) A–Concentration, B–DO, C–BOD, D–Sewage discharge

192. Which statement is/are true :-

- (a) The green house effect is a naturally occurring phenomenon
- (b) It is responsible for heating of earth surface and atmosphere
- (c) Without green house effect the average temperature at surface at earth would have been a chilling upto -18°C
- (d) Clouds and gases reflect about one fourth of the incoming solar radiation

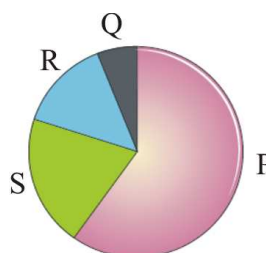
- (1) ac only
- (2) abc only
- (3) ab only
- (4) abcd

193. Match the column I with column II :-

	Column-I		Column-II
(i)	Water pollution Act	(a)	1986
(ii)	Wild protection life Act	(b)	1981
(iii)	Air pollution Act	(c)	1972
(iv)	NEPA	(d)	1974

	i	ii	iii	iv
(1)	c	b	a	d
(2)	d	c	b	a
(3)	a	c	b	d
(4)	a	b	c	d

194.



Above graph denotes the relative contribution of various green house gases to global warming. Identify P,Q,R,S

- (1) P = CO₂-60%; Q = CFC-14%;
R = N₂O-6%; S = CH₄-20%
- (2) P = CO₂-60%; Q = N₂O-6%;
R = CFC-14%; S = CH₄-20%
- (3) P = CO₂-60%; Q = CH₄- 6%;
R = CFC = 20%; S = N₂O-14%
- (4) P = CO₂-60%; Q = CFC-14%;
R = N₂O-6%; S = CH₄-30%

195. Match the Column-I with Column-II :-

	Column-I		Column-II
A	Kyoto protocol	i	Biodiversity conservation
B	Montreal protocol	ii	Climate change
C	Rio Earth summit	iii	Sustainable development
D	World summit	iv	Ozone protection
		v	Wetland conservation

- (1) A-ii, B-iv, C-iii, D-i
 (2) A-iv, B-ii, C-v, D-iii
 (3) A-iv, B-ii, C-v, D-i
 (4) A-ii, B-iv, C-i, D-iii

196. Identify the incorrect statement related to pollination-

- (1) Wind pollination is more common abiotic pollination
 (2) Flower produce foul odour to attract flies and beetles for pollination
 (3) Water pollination is quite rare abiotic pollination
 (4) Insects particularly bees are the least pollinating agent

197. A typical angiosperm the anther is ____

(A)____, (B)____ and (C)____.

	(A)	(B)	(C)
(1)	Pentagonal	Bilobed	Dithecous
(2)	Tetragonal	Monolobed	Monothecous
(3)	Tetragonal	Bilobed	Dithecous
(4)	Tetragonal	Bilobed	Tetrathecous

198. How many meiotic and mitotic division are required for the development of mature male gametophyte?

- (1) 1-Meiotic and 3-Mitotic
 (2) 1-Meiotic and 1-Mitotic
 (3) 1-Meiotic and 2-Mitotic
 (4) 2-Meiotic and 2-Mitotic

199. Match the Column-I with Column-II

	Column-I (Part)		Column-II (Ploidy)
(a)	Nucellus	(i)	(2n)
(b)	MMC (Megaspore Mother cell)	(ii)	(n)
(c)	Functional Megaspore	(iii)	(2n)
(d)	Embryo sac	(iv)	(n)

- (1) (a-i); (b-iii), (c-ii), (d-iv)
 (2) (a-ii); (b-i), (c-iv), (d-iii)
 (3) (a-iii); (b-iv), (c-i), (d-ii)
 (4) (a-iv); (b-iii), (c-ii), (d-i)

200. Which of the following is an example of water pollinated plant ?

- (1) Water hyacinth
 (2) Water Lily
 (3) *Vallisneria*
 (4) All of the above

SPACE FOR ROUGH WORK

 QUESTIONS BREAK-UP CHART FOR MINOR TEST FROM CURRENT & PREVIOUS UNITS (Session : 2023 - 2024)			
ENTHUSE ADV. COURSE : MED TG: @Chalnaayaaar			
MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
04 18-06-2023	SYLLABUS COVERED FROM 22-05-2023 TO 17-06-2023	P	ELECTROSTATICS
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